

Qualification Pack



Cold Chain Specialist (Green Engineering)

QP Code: LSC/Q9202

Version: 1.0

NSQF Level: 6

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LSC/Q9202: Cold Chain Specialist (Green Engineering)

Brief Job Description

The individual at work manages sustainability in cold chain by analyzing data related to energy use, identifying inefficiencies and implementing ways towards sustainability. The person is also responsible for maintaining equipment for temperature and storage conditions, undertaking water and effluent treatment in the plant and managing control systems of the refrigeration system components.

Personal Attributes

The job requires the individual to have high concentration for long periods of time, excellent vision, high stamina, good hand eye coordination and ability to work in cold temperatures for long duration.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [LSC/N9201: Implement energy efficiency and greening aspects in the cold storage](#)
2. [LSC/N9202: Oversee modified atmosphere requirements for the products](#)
3. [LSC/N9203: Undertake water treatment and conservation](#)
4. [LSC/N9204: Manage the renewed engineering system with greening aspects in cold storage](#)
5. [LSC/N9901: Maintain food and personnel safety, health and hygiene in cold storage plant](#)
6. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Logistics
Sub-Sector	Cold Chain Logistics Solutions
Occupation	Engineering, Sustainability
Country	India
NSQF Level	6
Credits	20

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/1324
Minimum Educational Qualification & Experience	Completed 4 year UG program (In Mechanical/ Electrical/ Electronics/ Refrigeration Engineering/ Food Technology/ Pharma/ Relevant trade)) with 1 Year of experience in cold storage operations OR Diploma (Completed 2-year diploma (after 12th Grade) in Mechanical/ Electrical/ Electronics/ Refrigeration Engineering/ Food Technology/ Pharma/ Relevant trade) with 3 Years of experience in cold storage operations OR Previous relevant Qualification of NSQF Level (5.5) with 2 Years of experience in cold storage operations OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience in cold storage operations
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	23 Years
Last Reviewed On	NA
Next Review Date	15/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-06-TW-02192-2024-V1-LSC
NQR Version	1.0

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LSC/N9201: Implement energy efficiency and greening aspects in the cold storage

Description

This OS unit is about analyzing energy usage in various processes of the cold chain, identifying areas to improve efficiency and undertake measures for the same.

Scope

The scope covers the following :

- Analyse data related to energy usage
- Identify inefficiencies in energy consumption and ways to fix them
- Implement ways to minimize energy inefficiencies

Elements and Performance Criteria

Analyse data related to energy usage

To be competent, the user/individual on the job must be able to:

- PC1.** identify parameters impacting energy usage
- PC2.** perform correlation between energy consumption and the parameters identified to impact energy usage
- PC3.** monitor electrical energy consumption, temperature, Relative Humidity (RH) and air pressure
- PC4.** define threshold for parameters in energy consumption
- PC5.** check all thermostat set points
- PC6.** evaluate costs and benefits of re-piping the facilities
- PC7.** perform thermographic inspection to analyse cold areas where there is poor insulation

Identify inefficiencies in energy consumption and ways to fix them

To be competent, the user/individual on the job must be able to:

- PC8.** separate energy consumption of refrigeration system from the energy used by the whole plant
- PC9.** identify the quantum of undercooling and overcooling during the analysis period
- PC10.** identify the impact of weather conditions on the energy consumed
- PC11.** explore ways to control lighting by daylight sensors and occupancy sensors
- PC12.** identify ways to adjust chiller or refrigeration equipment to achieve better performance
- PC13.** look for ways to promote renewable energy by utilizing bio-waste to generate biogas, wherever possible
- PC14.** improve evaporator performance by looking for ways to reduce fan motor horsepower

Implement ways to minimize energy inefficiencies

To be competent, the user/individual on the job must be able to:

- PC15.** use eco-friendly refrigerants with minimal global warming potential of ozone depleting substances

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- PC16.** ensure to pick air cooled condenser or evaporative condenser based on the refrigerant used, size of the system and availability of water
- PC17.** achieve optimal energy usage conditions for the chiller
- PC18.** improve part-load performance for evaporators, condensers and compressors
- PC19.** reduce refrigeration load by checking under-floor heating, insulation levels, warehouse doors usage
- PC20.** reduce load in lighting of the warehouse, by using high efficiency lighting (sodium lights or high frequency fluorescents)
- PC21.** choose energy-efficient lighting and recommend installing skylights, LEDs with sensors (intelligent occupancy sensors, motion sensors, heat maps), etc.
- PC22.** ensure that Lighting Power Density (LPD) does not exceed the required
- PC23.** recommend replacing smart appliances instead of traditional high power consuming standard devices, e.g., HVAC
- PC24.** use machinery and equipment like MHE that runs on electric or alternative clean energy
- PC25.** employ automatic refrigerant leak detection systems
- PC26.** reduce heat load by improving insulation and reducing air leakage
- PC27.** select low power consumed per ton of refrigeration, while upgrading evaporator units
- PC28.** upgrade to high efficiency condenser units, sensor-controlled ventilation systems etc.
- PC29.** use sliding doors instead of traditional freezer doors, which are better insulated, require low maintenance, reduce frost build up, thereby reducing overall energy consumption
- PC30.** use energy efficient PVC strip curtains for the cold storage doors or air curtains, to reduce air exchange during door openings
- PC31.** ensure that the floor heaters are working properly and well protected for the deep freezers

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** minimum acceptable standards for energy efficiency required by the company
- KU2.** corporate policy regarding methods of calculating contribution to global warming
- KU3.** how to undertake yearly energy audits on the plant
- KU4.** organizational policies and guidelines
- KU5.** reporting structure
- KU6.** communication with relevant people in the organization to take their buy-in for energy optimizing activities
- KU7.** refrigerant usage with its relevant safety and security procedures
- KU8.** procedures to follow during system emergency issues
- KU9.** roles and responsibilities of labourers in the cold storage area
- KU10.** how to measure electrical energy consumed, temperature, relative humidity and air pressure
- KU11.** how to regularly collect data regarding temperature set point, discharge and suction pressure, operating hours of the chiller etc
- KU12.** relation between carbon emissions and refrigerant leaks

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- KU13.** how to analyse what proportion of energy is used for each of the following - chiller/refrigeration, warehouse lighting, dock and freezer doors, battery charging, freezer floor heating, maintenance activities, storage and reefer vehicles
- KU14.** application of refrigeration for different types of products
- KU15.** characteristics of the products dealt with
- KU16.** cold storage and transport requirement conditions for products dealt with
- KU17.** various data acquisition systems available for plant refrigeration
- KU18.** the principles of cooling system design, carbon dioxide cascading, defrost system and purgers (for air and water)
- KU19.** the harmful effects of frost and ways to keep it out of the refrigerated spaces
- KU20.** types of refrigeration compressors available
- KU21.** how to perform thermal profile by collection of data from temperature sensors
- KU22.** how to record ambient conditions
- KU23.** how to alert in case of critical deviations in energy usage
- KU24.** chiller operations, compressor sequencing, and capacity regulation
- KU25.** how precooling of condenser air in air cooled system increases energy efficiency
- KU26.** collecting operational parameters for refrigeration equipment and/or chiller used
- KU27.** under-floor heating system to know how pump and heating is controlled
- KU28.** usage of occupancy sensor and daylight sensor
- KU29.** different techniques to control defrosting
- KU30.** selection of accessories and tools required for various activities
- KU31.** parameters of global sustainability
- KU32.** significance of CO₂ emissions and its impact on global warming
- KU33.** knowledge about material footprint per capita and material footprint per GDP
- KU34.** keep updated about the initiatives taken by the government and various organisations for sustainability
- KU35.** legislative requirements and organisational procedures for waste management and disposal
- KU36.** parameters and data required for material and energy audit

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read about energy consumption patterns by different processes in cold chain
- GS2.** read about energy efficiency improvement techniques
- GS3.** read safety instructions
- GS4.** note down energy consumption
- GS5.** prepare report on the efficacy of energy efficiency techniques employed
- GS6.** communicate to the workers, on the precautions to be taken regarding energy inefficiencies in the cold chain
- GS7.** interact with other employees to work efficiently
- GS8.** choose from multiple energy saving alternatives in the plant functioning

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- GS9.** decide on the priority of tasks planned
- GS10.** plan and organize incremental targets towards achieving energy efficiency
- GS11.** plan resources by selecting, training, and ensuring discipline amongst them
- GS12.** spot origins of process disruptions
- GS13.** re-schedule tasks in case of delays or requirements by other departments in the organization
- GS14.** interpret equipment and process diagrams to identify which components can work more efficiently
- GS15.** analyze relation between energy consumed and plant parameters
- GS16.** analyze time series data, regarding temperature, humidity or air pressure
- GS17.** analyze critical recurring issues and identify measures to solve the same
- GS18.** use acquired knowledge to trace inefficiencies in the process
- GS19.** plan and prioritize tasks based on the implications of the energy inefficiencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyse data related to energy usage</i>	6	12	-	3
PC1. identify parameters impacting energy usage	0.5	1	-	-
PC2. perform correlation between energy consumption and the parameters identified to impact energy usage	1	1	-	1
PC3. monitor electrical energy consumption, temperature, Relative Humidity (RH) and air pressure	1	2	-	-
PC4. define threshold for parameters in energy consumption	1	2	-	-
PC5. check all thermostat set points	0.5	2	-	-
PC6. evaluate costs and benefits of re-piping the facilities	1	2	-	1
PC7. perform thermographic inspection to analyse cold areas where there is poor insulation	1	2	-	1
<i>Identify inefficiencies in energy consumption and ways to fix them</i>	7	14	-	2
PC8. separate energy consumption of refrigeration system from the energy used by the whole plant	1	2	-	-
PC9. identify the quantum of undercooling and overcooling during the analysis period	1	2	-	-
PC10. identify the impact of weather conditions on the energy consumed	1	2	-	-
PC11. explore ways to control lighting by daylight sensors and occupancy sensors	1	2	-	1
PC12. identify ways to adjust chiller or refrigeration equipment to achieve better performance	1	2	-	-
PC13. look for ways to promote renewable energy by utilizing bio-waste to generate biogas, wherever possible	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. improve evaporator performance by looking for ways to reduce fan motor horsepower	1	2	-	1
<i>Implement ways to minimize energy inefficiencies</i>	17	34	-	5
PC15. use eco-friendly refrigerants with minimal global warming potential of ozone depleting substances	1	2	-	-
PC16. ensure to pick air cooled condenser or evaporative condenser based on the refrigerant used, size of the system and availability of water	1	2	-	-
PC17. achieve optimal energy usage conditions for the chiller	1	2	-	1
PC18. improve part-load performance for evaporators, condensers and compressors	1	2	-	1
PC19. reduce refrigeration load by checking under-floor heating, insulation levels, warehouse doors usage	1	2	-	-
PC20. reduce load in lighting of the warehouse, by using high efficiency lighting (sodium lights or high frequency fluorescents)	1	2	-	1
PC21. choose energy-efficient lighting and recommend installing skylights, LEDs with sensors (intelligent occupancy sensors, motion sensors, heat maps), etc.	1	2	-	-
PC22. ensure that Lighting Power Density (LPD) does not exceed the required	1	2	-	-
PC23. recommend replacing smart appliances instead of traditional high power consuming standard devices, e.g., HVAC	1	2	-	-
PC24. use machinery and equipment like MHE that runs on electric or alternative clean energy	1	2	-	-
PC25. employ automatic refrigerant leak detection systems	1	2	-	1
PC26. reduce heat load by improving insulation and reducing air leakage	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. select low power consumed per ton of refrigeration, while upgrading evaporator units	1	2	-	-
PC28. upgrade to high efficiency condenser units, sensor-controlled ventilation systems etc.	1	2	-	1
PC29. use sliding doors instead of traditional freezer doors, which are better insulated, require low maintenance, reduce frost build up, thereby reducing overall energy consumption	1	2	-	-
PC30. use energy efficient PVC strip curtains for the cold storage doors or air curtains, to reduce air exchange during door openings	1	2	-	-
PC31. ensure that the floor heaters are working properly and well protected for the deep freezers	1	2	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	LSC/N9201
NOS Name	Implement energy efficiency and greening aspects in the cold storage
Sector	Logistics
Sub-Sector	Cold Chain Logistics Solutions
Occupation	Engineering, Sustainability
NSQF Level	6
Credits	3
Version	2.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

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LSC/N9202: Oversee modified atmosphere requirements for the products

Description

This OS is about preparing, maintaining and monitoring cold room, container and equipment for modified atmosphere requirements of various products.

Scope

The scope covers the following :

- Prepare and monitor cold room or container for modified atmosphere storage
- Prepare and maintain equipment for modified atmosphere conditions
- Monitor and report modified atmosphere conditions

Elements and Performance Criteria

Prepare and monitor cold room or container for modified atmosphere storage

To be competent, the user/individual on the job must be able to:

- PC1.** maintain temperature ranges ideal for the products stored or transported
- PC2.** maintain concentration of oxygen, carbon dioxide, nitrogen and ethylene as per the commodities stored
- PC3.** consider product heat and set ventilation to control cooling and carbon dioxide level
- PC4.** monitor composition of gases regularly and accurately
- PC5.** monitor for chilling injury symptoms on the products
- PC6.** ensure to keep the room or container sealed
- PC7.** ensure proper sanitation to avoid conditions which favours thriving of pathogenic microorganisms

Prepare and maintain equipment for modified atmosphere conditions

To be competent, the user/individual on the job must be able to:

- PC8.** assign a resource to maintain records of refrigeration equipment
- PC9.** operate nitrogen generator with its controls and fan blowers
- PC10.** regularly sample air parameters in the cold chamber to check the conditions
- PC11.** monitor when to stop the system and replenish the cold chamber with fresh air when required
- PC12.** ensure to retrofit container with purge port assembly, when they contain perishable products and transported, to make it suitable for modified atmosphere use
- PC13.** prepare absorbers and adsorbers of oxygen, carbon dioxide, ethylene and water

Monitor and report modified atmosphere conditions

To be competent, the user/individual on the job must be able to:

- PC14.** ensure that legal requirements are followed while collecting, moving or
- PC15.** prepare documentation regarding modified atmospheric conditions maintained for different products

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- PC16.** ensure that maintenance records of relevant equipment and cleaning records in the modified atmosphere storage area are made
- PC17.** report any faults in the readings of required modified atmosphere requirements
- PC18.** quantify extended storability of the products dealt with

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies and guidelines
- KU2.** reporting structure
- KU3.** refrigerant usage with its relevant safety and security procedures
- KU4.** procedures to follow during emergency issues
- KU5.** roles and responsibilities of labourers in the cold storage area
- KU6.** accessories and tools required for maintaining modified atmosphere
- KU7.** typical ratios of different gases used in modified atmosphere requirements for different types of products
- KU8.** importance of having modified atmosphere to control shelf life of various products
- KU9.** different types of processes used to attain modified atmosphere requirements
- KU10.** different types of equipments required in modified atmosphere storage and packing
- KU11.** methods to alter atmosphere, like, controlled atmosphere, modified atmosphere, fresh air exchange, and packaging solutions
- KU12.** benefits and detrimental effects of modified atmosphere on products
- KU13.** storage requirements of various types of products
- KU14.** common commodities which require modified atmosphere
- KU15.** key physical and chemical stages in natural ripening of perishable products
- KU16.** how environmental conditions like cold room temperature and humidity affect life cycle of products
- KU17.** timescales of ripening and how to control them, in case of fresh produce ripening
- KU18.** how the condition of one product is affected by the presence of other products handled alongside in the cold chain
- KU19.** exposure of products to ethylene with regards to its respiration rate and ripening
- KU20.** methods for assessing product quality stored in modified atmosphere conditions
- KU21.** relation between product respiration rate and atmospheric composition
- KU22.** effects of modified atmospheric conditions on the skin colour, firmness, texture of the products
- KU23.** relation between metabolism rate and variations in temperature, relative humidity and oxygen levels
- KU24.** parameters of global sustainability
- KU25.** significance of CO₂ emissions and its impact on global warming
- KU26.** knowledge about material footprint per capita and material footprint per GDP
- KU27.** the initiatives taken by the government and various organisations for sustainability

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KU28. legislative requirements and organisational procedures for waste management and disposal

KU29. parameters and data required for material and energy audit

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** process required for different types of products
- GS2.** read equipment manual to handle its operation
- GS3.** read documents regarding modified atmosphere concepts
- GS4.** read safety instructions
- GS5.** write down process parameters in modified atmospheric conditions
- GS6.** write observations related to the process
- GS7.** communicate all the activities required to be done to the subordinates
- GS8.** listen to the queries and clarify doubts
- GS9.** communicate any issue that may arise in the modified atmosphere process to the concerned technical solver
- GS10.** interact frequently with other employees to work efficiently
- GS11.** decide on the priority of tasks
- GS12.** plan routine checks to ensure modified atmosphere conditions are maintained
- GS13.** organize tasks and allocate labour resources for the process to maintain modified atmosphere
- GS14.** organize activities based on budgeting constraints
- GS15.** enforce exact modified atmosphere storage and hygiene conditions as per the products handled for the organization or for its customers
- GS16.** solve any work related problems that workers may face
- GS17.** handle allotment of tasks to workers in case of staff shortage or delays in activities
- GS18.** interpret process flowcharts for modified atmosphere required for different products
- GS19.** analyze critical recurring issues and identify measures to solve the same
- GS20.** identify ways to minimize chilling symptoms seen on the products
- GS21.** use acquired knowledge to trace issues in the process
- GS22.** understand mathematical models relating respiration of various commodities and atmospheric composition

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare and monitor cold room or container for modified atmosphere storage</i>	14	22	-	3
PC1. maintain temperature ranges ideal for the products stored or transported	2	3	-	-
PC2. maintain concentration of oxygen, carbon dioxide, nitrogen and ethylene as per the commodities stored	2	3	-	-
PC3. consider product heat and set ventilation to control cooling and carbon dioxide level	2	3	-	-
PC4. monitor composition of gases regularly and accurately	2	3	-	-
PC5. monitor for chilling injury symptoms on the products	2	3	-	1
PC6. ensure to keep the room or container sealed	2	3	-	1
PC7. ensure proper sanitation to avoid conditions which favours thriving of pathogenic microorganisms	2	4	-	1
<i>Prepare and maintain equipment for modified atmosphere conditions</i>	11	20	-	4
PC8. assign a resource to maintain records of refrigeration equipment	2	3	-	-
PC9. operate nitrogen generator with its controls and fan blowers	2	3	-	1
PC10. regularly sample air parameters in the cold chamber to check the conditions	2	3	-	-
PC11. monitor when to stop the system and replenish the cold chamber with fresh air when required	2	3	-	1
PC12. ensure to retrofit container with purge port assembly, when they contain perishable products and transported, to make it suitable for modified atmosphere use	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. prepare absorbers and adsorbers of oxygen, carbon dioxide, ethylene and water	1	4	-	1
<i>Monitor and report modified atmosphere conditions</i>	5	18	-	3
PC14. ensure that legal requirements are followed while collecting, moving or	1	4	-	1
PC15. prepare documentation regarding modified atmospheric conditions maintained for different products	1	4	-	1
PC16. ensure that maintenance records of relevant equipment and cleaning records in the modified atmosphere storage area are made	1	4	-	-
PC17. report any faults in the readings of required modified atmosphere requirements	1	3	-	1
PC18. quantify extended storability of the products dealt with	1	3	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	LSC/N9202
NOS Name	Oversee modified atmosphere requirements for the products
Sector	Logistics
Sub-Sector	Cold Chain Logistics Solutions
Occupation	Engineering, Sustainability
NSQF Level	6
Credits	3
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

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LSC/N9203: Undertake water treatment and conservation

Description

This OS unit is about undertaking and monitoring water and effluent treatment program in the plant

Scope

The scope covers the following :

- Undertake water and effluent treatment program in the plant
- Monitor performance of the water and effluent treatment
- Implement water conservation

Elements and Performance Criteria

Undertake water and effluent treatment program in the plant

To be competent, the user/individual on the job must be able to:

- PC1.** ensure that the water treatment programme consists of controlling water circulation, chemical feed and routine monitoring
- PC2.** ensure that treated water is used on the product, to avoid the threat of external pathogens spoiling them
- PC3.** verify water treatment system compatibility with the plant operations
- PC4.** explore options to collect water from evaporator units, if they have defrost unit, and provide them to condenser or cooling tower
- PC5.** ensure to maintain condenser tubes clean after water treatment
- PC6.** identify treated effluent and tweak its treatment, if it is not as per the specification

Monitor performance of the water and effluent treatment

To be competent, the user/individual on the job must be able to:

- PC7.** record information and document necessary details for generating programme efficacy report
- PC8.** monitor remedial engineering works
- PC9.** confirm that the treatment areas are hygienic and safe
- PC10.** investigate faults and errors and take corrective actions
- PC11.** train operators for water and effluent treatment procedures employed
- PC12.** monitor overfeed or underfeed of water treatment chemicals
- PC13.** inspect chemical feed system to check for leaks

Implement water conservation

To be competent, the user/individual on the job must be able to:

- PC14.** find cost effective ways for treatment and recycle of COD (Chemical Oxygen Demand) wastewater
- PC15.** explore possibility of rainwater harvesting in case there is large roof surfaces for the cold storage plant

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PC16. develop processes to maintain and have zero wastage/ emission or net-positive water footprint by implementing ways to conserve and reuse water, e.g., installing automated/sensor-enabled water faucets, integrating sensors with IoT for transmission, installing low-flow faucets and toilets, fixing leaks promptly, optimising water-intensive processes, installing sewage water treatment plants etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of having water and effluent treatment programme in accordance with the operating procedures of the company
- KU2.** compliance and site specific requirements of the water and effluent treatment programme that should be in accordance with
- KU3.** how to wear protective gear while handling or overseeing water and effluent treatment
- KU4.** corrective actions that can be taken in accordance with company procedures
- KU5.** organizational policies and guidelines
- KU6.** reporting structure
- KU7.** refrigerant usage with its relevant safety and security procedures
- KU8.** safety procedures in using chemicals for the treatment
- KU9.** procedures to follow during emergency maintenance and repair issues
- KU10.** procedures for safe transport and disposal of waste materials
- KU11.** importance of effective water treatment for efficient and reliable refrigeration system operation
- KU12.** water related problems that can happen in the cold store plant, like corrosion and microbiological growth
- KU13.** how to make a risk assessment for work tasks in the cold chain where water or effluent is generated
- KU14.** different types of treatment systems available for water waste and effluent
- KU15.** concepts of microbiology and water chemistry which applies to effluent treatment
- KU16.** how to do legionella (bacteria) risk assessment
- KU17.** how to take measures to prevent cross contamination of various products dealt with in the cold chain
- KU18.** how to plan and record performance monitoring for the programme
- KU19.** how to determine feasible recovery rate for recycling of water
- KU20.** primary and secondary treatment methods of effluent waters
- KU21.** application of refrigeration for different types of products
- KU22.** usage of ozone in tertiary treatment to remove odour, colour and micro organisms in sewage water
- KU23.** harmful effects of overfeeding and underfeeding water treatment chemicals
- KU24.** methods of treating wastewater and recycling of water
- KU25.** parameters of global sustainability
- KU26.** significance of CO2 emissions and its impact on global warming

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- KU27.** knowledge about material footprint per capita and material footprint per GDP
- KU28.** the initiatives taken by the government and various organisations for sustainability
- KU29.** legislative requirements and organisational procedures for waste management and disposal
- KU30.** parameters and data required for material and energy audit

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read water and effluent treatment standards and procedures
- GS2.** read safety instructions
- GS3.** how to make documentation regarding water and effluent treatment activities
- GS4.** communicate to workers clearly about the requirements in water and effluent treatment activities
- GS5.** interact frequently with other employees to work efficiently
- GS6.** mentor all the workers under supervision to increase operational effectiveness
- GS7.** decide on the extent of water and effluent treatment done, based on the hygiene requirements and the products
- GS8.** plan maintenance activities as per standard requirements and manufacturer's instructions
- GS9.** plan equipment service based on its working condition
- GS10.** enforce storage and hygiene conditions as per the products handled for the organization or for its customers
- GS11.** ensure to solve operating problems as per best practices
- GS12.** re-schedule tasks in case of delays or requirements by other departments in the organization
- GS13.** interpret schematic diagrams of the treatment procedures
- GS14.** analyze efficacy of the treatment programme employed
- GS15.** find ways to reduce sewer costs
- GS16.** analyze critical recurring issues and identify measures to solve the same
- GS17.** interpret outputs from the water and effluent treatment programs applied
- GS18.** use acquired knowledge to trace faults in the process

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Undertake water and effluent treatment program in the plant</i>	10	29	-	4
PC1. ensure that the water treatment programme consists of controlling water circulation, chemical feed and routine monitoring	1	5	-	-
PC2. ensure that treated water is used on the product, to avoid the threat of external pathogens spoiling them	1	5	-	-
PC3. verify water treatment system compatibility with the plant operations	2	5	-	1
PC4. explore options to collect water from evaporator units, if they have defrost unit, and provide them to condenser or cooling tower	2	5	-	1
PC5. ensure to maintain condenser tubes clean after water treatment	2	5	-	1
PC6. identify treated effluent and tweak its treatment, if it is not as per the specification	2	5	-	1
<i>Monitor performance of the water and effluent treatment</i>	14	28	-	5
PC7. record information and document necessary details for generating programme efficacy report	2	4	-	1
PC8. monitor remedial engineering works	2	4	-	-
PC9. confirm that the treatment areas are hygienic and safe	2	4	-	1
PC10. investigate faults and errors and take corrective actions	2	4	-	-
PC11. train operators for water and effluent treatment procedures employed	2	4	-	1
PC12. monitor overfeed or underfeed of water treatment chemicals	2	4	-	1
PC13. inspect chemical feed system to check for leaks	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implement water conservation</i>	6	3	-	1
PC14. find cost effective ways for treatment and recycle of COD (Chemical Oxygen Demand) wastewater	2	1	-	-
PC15. explore possibility of rainwater harvesting in case there is large roof surfaces for the cold storage plant	2	1	-	1
PC16. develop processes to maintain and have zero wastage/ emission or net-positive water footprint by implementing ways to conserve and reuse water, e.g., installing automated/sensor-enabled water faucets, integrating sensors with IoT for transmission, installing low-flow faucets and toilets, fixing leaks promptly, optimising water-intensive processes, installing sewage water treatment plants etc.	2	1	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LSC/N9203
NOS Name	Undertake water treatment and conservation
Sector	Logistics
Sub-Sector	Cold Chain Logistics Solutions
Occupation	Engineering, Sustainability
NSQF Level	6
Credits	3
Version	2.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

LSC/N9204: Manage the renewed engineering system with greening aspects in cold storage

Description

This OS unit is about managing engineering system for the cold chain.

Scope

The scope covers the following :

- Manage control system for the main components of refrigeration system
- Identify the need for revamping existing facilities for energy saving
- Prepare documentation regarding management of engineering system

Elements and Performance Criteria

Manage control system for the main components of refrigeration system

To be competent, the user/individual on the job must be able to:

- PC1.** automate the operation of evaporators, condensers and compressors as much as possible
- PC2.** minimize the power consumption for evaporators, condensers and compressors combined
- PC3.** control evaporators when the cold room has attained desired temperature
- PC4.** ensure that hydraulic oil is adapted to the temperatures operated at the cold store

Identify the need for revamping existing facilities for energy saving

To be competent, the user/individual on the job must be able to:

- PC5.** identify different components of engineering system employed which needs to be checked at regular intervals
- PC6.** plan and schedule the frequency with which each component in the facility needs to be examined
- PC7.** analyse pressure drop between condenser and evaporator, to check if it crosses two pounds
- PC8.** examine insulated pipes which show signs of corrosion or frost appearances
- PC9.** examine corroded parts to analyse and decide regarding the need for re-insulation or maintenance

Prepare documentation regarding management of engineering system

To be competent, the user/individual on the job must be able to:

- PC10.** record information and document regarding operations of evaporators, condensers and compressors
- PC11.** document power consumption of refrigeration system at regular intervals
- PC12.** record observations made in the analysis of existing facilities and the needs identified for its revamp

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** organizational policies and guidelines
- KU2.** reporting structure
- KU3.** refrigerant usage with its relevant safety and security procedures
- KU4.** procedures to follow during emergency system issues
- KU5.** roles and responsibilities of labourers in the cold storage area
- KU6.** control system handling procedures for refrigeration equipment
- KU7.** procedures for safe transport and disposal of waste materials
- KU8.** cost and energy consumption of various equipment used in the organization
- KU9.** importance of control method for refrigeration system
- KU10.** mechanism of simple vapour compression refrigeration system and air refrigeration system
- KU11.** types of evaporators and factors affecting its heat transfer capacity
- KU12.** types of compressors, and the parameters for its working, like discharge pressure, suction pressure, compression ratio, compressor capacity and volumetric efficiency
- KU13.** types and working of condensers, and factors affecting its capacity
- KU14.** application of refrigeration for different types of products
- KU15.** working of hand operated expansion valve
- KU16.** different process freezing techniques like contact freezing and air blast freezing
- KU17.** that a facility needs a re-piping if pressure drop between condenser and evaporator crosses two pounds
- KU18.** how to check pressure drop by using pressure gauges
- KU19.** types, properties and thickness of insulation requirements
- KU20.** parameters of global sustainability
- KU21.** significance of CO₂ emissions and its impact on global warming
- KU22.** knowledge about material footprint per capita and material footprint per GDP
- KU23.** the initiatives taken by the government and various organisations for sustainability
- KU24.** legislative requirements and organisational procedures for waste management and disposal
- KU25.** parameters and data required for material and energy audit

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read about equipment engineering and understand its working
- GS2.** read and comprehend technical documents regarding mechanics and hydraulic aspects of cold chain engineering
- GS3.** read safety instructions
- GS4.** note down problems in the system
- GS5.** write down process notes regarding observations in mechanics and hydraulics of all equipments used
- GS6.** make analytical document or report regarding existing engineering systems
- GS7.** communicate and seek inputs from workers regarding working of various components
- GS8.** communicate to workers clearly about the requirements in managing control systems

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- GS9.** communicate technical knowledge regarding equipment functioning
- GS10.** listen and understand all queries or doubts the workers have in mechanics or hydraulic aspects of cold chain engineering
- GS11.** resolve emergency situations in system functioning
- GS12.** plan training activities
- GS13.** organize practical learning techniques
- GS14.** plan maintenance activities as per standard requirements and manufacturer's instructions
- GS15.** plan equipment service based on its working condition
- GS16.** help solve all queries or difficulties faced by trainees
- GS17.** re-schedule tasks in case of delays or requirements by other departments in the organization
- GS18.** handle allotment of tasks to workers in case of staff shortage or delays in activities
- GS19.** interpret equipment diagram and information to identify how each component is engineered to work
- GS20.** analyze critical recurring issues and identify measures to solve the same
- GS21.** assess working condition of various equipments
- GS22.** plan activities to minimize effects on normal working of the organization
- GS23.** optimize working of control systems in refrigeration equipment
- GS24.** use acquired knowledge to trace technical errors in the plant

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage control system for the main components of refrigeration system</i>	12	20	-	3
PC1. automate the operation of evaporators, condensers and compressors as much as possible	3	5	-	1
PC2. minimize the power consumption for evaporators, condensers and compressors combined	3	5	-	-
PC3. control evaporators when the cold room has attained desired temperature	3	5	-	1
PC4. ensure that hydraulic oil is adapted to the temperatures operated at the cold store	3	5	-	1
<i>Identify the need for revamping existing facilities for energy saving</i>	12	25	-	4
PC5. identify different components of engineering system employed which needs to be checked at regular intervals	3	5	-	-
PC6. plan and schedule the frequency with which each component in the facility needs to be examined	3	5	-	1
PC7. analyse pressure drop between condenser and evaporator, to check if it crosses two pounds	2	5	-	1
PC8. examine insulated pipes which show signs of corrosion or frost appearances	2	5	-	1
PC9. examine corroded parts to analyse and decide regarding the need for re-insulation or maintenance	2	5	-	1
<i>Prepare documentation regarding management of engineering system</i>	6	15	-	3
PC10. record information and document regarding operations of evaporators, condensers and compressors	2	5	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. document power consumption of refrigeration system at regular intervals	2	5	-	1
PC12. record observations made in the analysis of existing facilities and the needs identified for its revamp	2	5	-	1
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LSC/N9204
NOS Name	Manage the renewed engineering system with greening aspects in cold storage
Sector	Logistics
Sub-Sector	Cold Chain Logistics Solutions
Occupation	Engineering, Sustainability
NSQF Level	6
Credits	3
Version	2.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

LSC/N9901: Maintain food and personnel safety, health and hygiene in cold storage plant

Description

This OS unit is about complying with safety, health and hygiene at the workplace to have a hazard-free environment and avoid downtime

Scope

The scope covers the following :

- Take precautionary measures to avoid work hazards
- Follow standard health, safety and hygiene procedures

Elements and Performance Criteria

Take precautionary measures to avoid work hazards

To be competent, the user/individual on the job must be able to:

- PC1.** assess the various health, safety and environmental hazards in the cold storage
- PC2.** take necessary steps to eliminate or minimize the hazards
- PC3.** analyze the causes of accidents at the workplace
- PC4.** take preventive measures to avoid risk of cold burns and other injury due to contact with hot surfaces, gas, fire, hot fluids/ liquids, etc.
- PC5.** ensure the employees have access to first aid kit when needed
- PC6.** ensure to use personal protective equipment and safety gear such as gloves, jacket, footwear etc. for loading and unloading material in cold rooms to protect themselves from hypothermia, frostbite etc
- PC7.** ensure to display safety signs at places where necessary for people to be cautious
- PC8.** use rubber mats in the places where floors are constantly wet
- PC9.** ensure electrical precautions such as insulated clothing, adequate equipment insulation, dry work area, switch off the power supply when not required, etc
- PC10.** display emergency exit plan at prominent places and have emergency assembly area earmarked as a grid for easy counting of on duty associates and workers.
- PC11.** unplug the control panel, compressor, condensor etc before performing maintenance
- PC12.** report to the superior on any problems and hazards identified
- PC13.** install fire alarms (electrical/manual) in cold store/deep freeze and keep other safety devices like hammer/mallet in the storage area

Follow standard health, safety and hygiene procedures

To be competent, the user/individual on the job must be able to:

- PC14.** maintain appropriate ventilation in the cold rooms to avoid unacceptable accumulation of heat, condensation or odours
- PC15.** check and review the cold storage areas frequently

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- PC16.** stack items in an organized way and use safe lifting techniques to reduce risk of injuries from handling procedures at the storage areas
- PC17.** ensure no sign of pest infestation and install rodent traps, fly glues and insectocutors wherever needed
- PC18.** follow hygiene & sanitation standards of Government bodies like FSSAI, APEDA and /or EIA or importing countries like FAO, EU standards
- PC19.** use effective loading and unloading systems
- PC20.** proper stock rotation (First in First out) to be practiced
- PC21.** segregate damaged/non-conforming products from other products to designate area for appropriate disposition
- PC22.** fumigate containers depending upon product and contamination or as per customers requirement
- PC23.** avoid smoking, spitting, eating etc near food storage area
- PC24.** ensure reefers are covered, clean, free from pest infestation & other contaminants
- PC25.** dispose cold storage plant waste in the designated areas safely as per companys policies and rules
- PC26.** ensure to be safe while handling machines (generator, compressor, condenser etc.), gas (ammonia) and chemicals (ethylene, refrigerants etc.)
- PC27.** keep the floors free from oil, water and grease to avoid slippery surface
- PC28.** cut nails regularly and avoid applying nail paint. Avoid wearing bangles, rings, and chains in cold storage
- PC29.** wash hands with soap solution and dry under a dryer as they enter for duty or after using wash room
- PC30.** periodic examination of protective devices, pressure vessels and pipelines, and parts of pipework by a competent person to prevent defect that may give rise to danger
- PC31.** ensure workers suffering from abscess, boils etc. should be relieved from food handling
- PC32.** develop personal hygiene habits like brushing teeth, taking shower every day, wearing clean and tidy clothes after ironing etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's HR policies on personnel management
- KU2.** company's reporting structure
- KU3.** occupational health and safety standards
- KU4.** cold storage plant inspection checklist
- KU5.** companys sanitary standard operating procedures
- KU6.** procedures to follow during emergency maintenance issues
- KU7.** technical standards for design and construction of cold storages: Bureau of Indian standards (BIS), International standard (ISO) etc.
- KU8.** the purpose and usage of protective gears such as gloves , jackets etc. while working
- KU9.** use of first aid at workplace

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- KU10.** cold storage order, 1980
- KU11.** food safety and standards act, 2006
- KU12.** reporting procedure or hierarchy for signs of damage and potential hazards
- KU13.** methods to minimize accidental risks
- KU14.** safe storage and handling of chemicals like refrigerants, ammonia, ethylene etc.
- KU15.** loading and unloading systems
- KU16.** standard operating procedure for safety drills and equipment maintenance
- KU17.** operation of machines: compressor, condenser, evaporator etc.
- KU18.** emergency procedures to be followed in case of an mishap such as fire, accidents, etc. and communication of safety instructions to subordinate staff
- KU19.** emergency responses in case of malfunctioning of refrigeration equipment as a whole or its components like evaporator, condenser or compressor
- KU20.** solid, liquid and gaseous waste disposal, treatment norms and equipment
- KU21.** necessary action to be taken for the hazards identified
- KU22.** knowledge of Quality systems like BRC, FSSAI, ISO, FSSC, HACCP etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the relevant organisation policies, procedures and diagrams that identify health, safety and safe environmental practices.
- GS2.** read job sheets, company policy documents and information displayed at the workplace for health, safety and environment.
- GS3.** read notes/comments from the senior
- GS4.** fill up documentation related to health, safety and environmental standards, if required
- GS5.** verbally report health, safety and environmental hazards and poor organisation practice.
- GS6.** communicate to the supervisor about the work health, safety and environmental issues
- GS7.** receive instructions from supervisor on minimizing the risks
- GS8.** communicate with co-workers about the precautions to be taken for hazards free work
- GS9.** take preventive measures for the identified hazards
- GS10.** select appropriate hand tools and personal protection equipment
- GS11.** identify first aid needs in case of an injury
- GS12.** formalize and display evacuation plan at strategic locations
- GS13.** ensure targeted product delivery by practicing stipulated standards of occupational health safety and environmental measures
- GS14.** take care of personal and equipment protection
- GS15.** identify the hazards and suggest possible solutions
- GS16.** use safety equipment such as fire extinguisher during fire accidents
- GS17.** store tools in a safe way
- GS18.** analyse the seriousness of the hazards
- GS19.** evolve smooth workflow by avoiding hazards at workplace

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- GS20.** evaluate and apply the possible solutions for the hazards, as necessary
- GS21.** take care of personal and equipment protection
- GS22.** identify the hazards and suggest possible solutions
- GS23.** use safety equipment such as fire extinguisher during fire accidents
- GS24.** store tools in a safe way
- GS25.** analyse the seriousness of the hazards
- GS26.** evolve smooth workflow by avoiding hazards at workplace
- GS27.** evaluate and apply the possible solutions for the hazards, as necessary

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Take precautionary measures to avoid work hazards</i>	12	23	-	6
PC1. assess the various health, safety and environmental hazards in the cold storage	0.5	1	-	-
PC2. take necessary steps to eliminate or minimize the hazards	1	2	-	1
PC3. analyze the causes of accidents at the workplace	0.5	1	-	-
PC4. take preventive measures to avoid risk of cold burns and other injury due to contact with hot surfaces, gas, fire, hot fluids/ liquids, etc.	1	2	-	1
PC5. ensure the employees have access to first aid kit when needed	1	2	-	-
PC6. ensure to use personal protective equipment and safety gear such as gloves, jacket, footwear etc. for loading and unloading material in cold rooms to protect themselves from hypothermia, frostbite etc	1	2	-	1
PC7. ensure to display safety signs at places where necessary for people to be cautious	1	2	-	-
PC8. use rubber mats in the places where floors are constantly wet	0.5	2	-	1
PC9. ensure electrical precautions such as insulated clothing, adequate equipment insulation, dry work area, switch off the power supply when not required, etc	2	2	-	-
PC10. display emergency exit plan at prominent places and have emergency assembly area earmarked as a grid for easy counting of on duty associates and workers.	1	2	-	-
PC11. unplug the control panel, compressor, condensor etc before performing maintenance	0.5	1	-	-
PC12. report to the superior on any problems and hazards identified	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. install fire alarms (electrical/manual) in cold store/deep freeze and keep other safety devices like hammer/mallet in the storage area	1	2	-	1
<i>Follow standard health, safety and hygiene procedures</i>	18	37	-	4
PC14. maintain appropriate ventilation in the cold rooms to avoid unacceptable accumulation of heat, condensation or odours	0.5	1	-	-
PC15. check and review the cold storage areas frequently	0.5	2	-	-
PC16. stack items in an organized way and use safe lifting techniques to reduce risk of injuries from handling procedures at the storage areas	1	2	-	-
PC17. ensure no sign of pest infestation and install rodent traps, fly glues and insectocutors wherever needed	1	2	-	1
PC18. follow hygiene & sanitation standards of Government bodies like FSSAI, APEDA and /or EIA or importing countries like FAO, EU standards	1	2	-	-
PC19. use effective loading and unloading systems	1	2	-	-
PC20. proper stock rotation (First in First out) to be practiced	1	2	-	-
PC21. segregate damaged/non-conforming products from other products to designate area for appropriate disposition	1	2	-	1
PC22. fumigate containers depending upon product and contamination or as per customers requirement	1	2	-	-
PC23. avoid smoking, spitting, eating etc near food storage area	1	2	-	-
PC24. ensure reefers are covered, clean, free from pest infestation & other contaminants	1	2	-	1
PC25. dispose cold storage plant waste in the designated areas safely as per companys policies and rules	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. ensure to be safe while handling machines (generator, compressor, condenser etc.), gas (ammonia) and chemicals (ethylene, refrigerants etc.)	1	2	-	1
PC27. keep the floors free from oil, water and grease to avoid slippery surface	1	2	-	-
PC28. cut nails regularly and avoid applying nail paint. Avoid wearing bangles, rings, and chains in cold storage	1	2	-	-
PC29. wash hands with soap solution and dry under a dryer as they enter for duty or after using wash room	1	2	-	-
PC30. periodic examination of protective devices, pressure vessels and pipelines, and parts of pipework by a competent person to prevent defect that may give rise to danger	1	2	-	-
PC31. ensure workers suffering from abscess, boils etc. should be relieved from food handling	1	2	-	-
PC32. develop personal hygiene habits like brushing teeth, taking shower every day, wearing clean and tidy clothes after ironing etc.	1	2	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LSC/N9901
NOS Name	Maintain food and personnel safety, health and hygiene in cold storage plant
Sector	Logistics
Sub-Sector	Cold Chain Logistics Solutions
Occupation	Cold Chain Operations
NSQF Level	5.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion.
5. To pass the Qualification Pack, every trainee should score a minimum of 70% for NSQF level 4 & above job roles and 50% for NSQF level 1 to 3 job roles.
6. In case of unsuccessful completion, the trainee may seek re-assessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LSC/N9201.Implement energy efficiency and greening aspects in the cold storage	30	60	-	10	100	20
LSC/N9202.Oversee modified atmosphere requirements for the products	30	60	-	10	100	15
LSC/N9203.Undertake water treatment and conservation	30	60	-	10	100	20
LSC/N9204.Manage the renewed engineering system with greening aspects in cold storage	30	60	-	10	100	15
LSC/N9901.Maintain food and personnel safety, health and hygiene in cold storage plant	30	60	-	10	100	15
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	15
Total	170	330	-	50	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.